



Predatory crows use behavioural flexibility to overcome a novel, toxic prey

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Abstract

Some invasive species threaten native species because they possess traits to which natives are evolutionarily naïve. In Australia, many predators have been severely impacted by the introduction of cane toads, which produce toxins that native species have never encountered. Torresian crows, however, have been observed consuming cane toads safely by avoiding their toxin glands, suggesting learned behavioural adaptations. To investigate this, we used baited camera stations to test whether wild crows with 1–10 years exposure to cane toads consumed toads differently compared to non-toxic native frogs. Crows were presented with four prey types: native frogs, intact toads, toads with toxin glands removed, and toads with toxin glands removed but toxin reapplied. In all trials with native frogs and toads lacking toxin glands, crows consumed the entire prey item, correctly recognising them as non-toxic. With intact toads, crows consumed only non-toxic parts in 89% of trials, demonstrating strategies to safely exploit this toxic prey. When toxin glands were removed but toxin reapplied, crows consumed only non-toxic parts in 58% of trials, suggesting that some individuals may rely on the presence of toxin glands as a visual cue of toxicity. These findings provide clear evidence of behavioural flexibility in Torresian crows, highlighting their capacity to rapidly adapt to novel threats posed by invasive species.

Significance statement

Our study provides experimental evidence that a native predator can rapidly adjust feeding behaviours to safely exploit a toxic invasive species. Torresian crows in northern Australia demonstrated the ability to discriminate between toxic and non-toxic prey and to modify their handling strategies to avoid toxic regions of cane toads. This behavioural flexibility offers a mechanism by which native predators may persist despite the severe ecological impacts of the cane toad invasion. By identifying the cues crows use to assess prey toxicity, our findings contribute to a broader understanding of how behavioural flexibility and learning shape predator–prey interactions in novel environments. More broadly, this work highlights the role of behavioural flexibility in mediating species responses to rapid environmental change, with implications for invasion biology, adaptation, and the resilience of native fauna.

Keywords Torresian crow · *Corvus orru* · Cane toad · *Rhinella marina* · Predator-prey interaction

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Introduction

Humans are causing unprecedented levels of environmental disturbance, with the scale and speed of these impacts far exceeding rates of change prior to human influence. The severity and rapidity of such changes mean that species often have insufficient time to adapt through natural selection, leading to population declines and extinctions (Sih et al. 2011; Gruber et al. 2019). However, some species can cope with rapid change through behavioural flexibility – the ability to quickly modify an existing behaviour or develop a new one in response to novel stressors or environmental

change (Sol et al. 2005; Tuomainen and Candolin 2011; Gruber et al. 2019; Lea et al. 2020). This flexibility enables animals to deal with unfamiliar stimuli and even exploit novel resources within remarkably short timeframes (Sih et al. 2011; Ruland and Jeschke 2020).

One increasingly common environmental challenge is the spread of invasive species, which can introduce novel traits and pressures that native species have not previously encountered (Vitousek et al. 1996; Prentis et al. 2008; Greenlees et al. 2010; Tuomainen and Candolin 2011). These pressures include predation, competition for resources, and ecosystem disruption (Gurevitch and Padilla 2004). The extent of an invader's impact depends both on its traits and the ability of native species to counter those traits. In this context, behavioural flexibility is advantageous, potentially allowing native species to rapidly adapt to novel invader traits. While behavioural flexibility is frequently cited as a key factor in the success of invasive species, examples of native species adapting their behaviour to mitigate invasive impacts are less well documented. A few studies have reported native animals altering foraging behaviour (e.g. Chow et al. 2018), and prey handling strategies (e.g. Langkilde 2009) to cope with invasive species, but such cases remain rare.

In Australia, the invasive cane toad (*Rhinella marina*) is a highly successful invader that has had severe impacts on native fauna (Doody et al. 2006, 2014; Smith and Phillips 2006; Shine 2010; Pettit et al. 2021b) due to lethal toxins in the parotoid glands (enlarged glands behind the head) and dorsal skin glands (Phillips et al. 2003; Smith and Phillips 2006; Shine 2010, 2014). Predators in the toads' native range have had time to evolve resistance to these toxins (Callaway and Ridenour 2004; Smith and Phillips 2006; Vilcinskas 2015). In contrast, the introduction and rapid spread of cane toads in Australia means that native predators are evolutionarily naïve to the toxins and highly susceptible to lethal poisoning. This has caused population declines and local extinctions in more than 90% of some frog-eating predators, including yellow spotted monitors (*Varanus panoptes*) and northern quolls (*Dasyurus hallucatus*) (Shine 2010; Doody et al. 2015, 2017, 2021; Pettit et al. 2021a). However, some native predators, including birds such as crows and kites, and some rodents, have been observed safely consuming invasive cane toads by avoiding the toxic glands (Donato and Potts 2004; Beckmann and Shine 2011; Parrott et al. 2019). This has led to the hypothesis that predators with high behavioural flexibility or innovation can rapidly learn strategies to consume cane toads safely, thereby avoiding the severe population declines seen in less adaptable species (Donato and Potts 2004; Parrott et al. 2019; Clarke et al. 2020; Pettit et al. 2021a).

The corvid family, including crows, ravens, and rooks, is renowned for advanced cognition, behavioural flexibility,

and innovative problem solving (Hunt 1996; Kenward et al. 2006; Taylor et al. 2010; Auersperg et al. 2011). The Torresian crow (*Corvus orru*) is widely distributed across northern Australia, completely overlapping the current range of the invasive cane toad. Anecdotal reports suggest that Torresian crows have rapidly adapted to this novel prey by flipping toads and consuming only the non-toxic tissues, such as internal organs and muscle (Donato and Potts 2004; Franklin 2005; Beckmann and Shine 2009; Pettit et al. 2021b). This makes Torresian crows an excellent model for investigating the role of behavioural flexibility in mitigating the risks posed by a highly toxic invader. To date, reports of this behaviour are limited to opportunistic observations, underscoring the need for empirical investigation. Here, we address this gap by experimentally presenting crows with different prey items, predicting that, if crows exhibit behavioural flexibility in dealing with toxic cane toads, they should distinguish between harmless native frogs and toxic toads, consuming most of or all of a frog while avoiding the toad's toxic regions. Additionally, we aimed to identify whether crows rely on visual cues, chemical cues, or both, to discriminate between toxic and non-toxic prey.

Methods

Study species

Torresian crows are non-migratory corvid species found across tropical northern Australia. They are diurnal, generalist foragers that frequently scavenge along roads where carrion from vehicle strikes is common (Clifton and Jones 2017; Bragato et al. 2022). This behaviour brings them into contact with cane toads, a largely nocturnal species often found dead on roads. Encounters with live toads are less frequent, although cane toads can sometimes be active during the day (Doody et al. 2019). Cane toad toxins do not readily breakdown, persisting in carcasses for several days to months (Crossland et al. 2011), meaning that crows are frequently exposed to toxic cane toad carrion.

Study area

We set-up 153 baited camera stations at 24 unique sites across a six-week period (June–July 2019) in the East Kimberley region of Western Australia, an area invaded by cane toads. Although exact invasion dates are unknown, we estimated cane toads had been present at the study sites for between 1 and 10 years based on Western Australian government cane toad distribution maps (Department of Biodiversity Conservation and Attractions (DBCA) 2024). Our sites spanned approximately 700 km from Kununurra

(15°41'S, 128°43'E) to Fitzroy Crossing (15°41'S, 128°43'E). We selected sites haphazardly along the Great Northern Highway and Gibb River Road focusing on accessible townships or campgrounds where crows were common (Table S1, Fig. S1). Sites within the same township were at least 5 km apart, with most separated by 20 km or more to ensure sampling of distinct crow populations and social groups. Within each site, we targeted landfills or campgrounds to maximise the likelihood of crow encounters at feeding stations.

Feeding stations

We presented crows with dead prey items on baited feeding stations (Fig. S2) with a GoPro camera to record the encounters. To secure the prey, we created a small puncture hole beneath the spine using a hollow 15-gauge microchip needle, then threaded fishing line (ROVEX® 45 lb, 0.6 mm continuing onto Neptune Tackle 50 lb, 0.6 mm) through the puncture. We knotted the fishing line twice to hold the prey securely and then attached the fishing line to an eyelet at the centre of a wooden board, leaving 10 cm of slack. This setup prevented crows from removing the prey item from the camera's field of view while allowing crows to manipulate the prey item. Feeding stations were left for a 2-hour period with cameras recording continuously.

Prey item treatments

We presented crows with one of four prey items: (1) a dead adult native frog ('native frog'), to establish baseline feeding behaviour on non-toxic anurans; (2) an intact dead juvenile cane toad ('intact toad'). We used juvenile rather than adult cane toads because these were closer in size to adult native frogs; (3) a dead juvenile cane toad with the parotoid glands removed ('toad no glands'); and, (4) a dead juvenile cane toad with parotoid glands removed but toxins smeared back onto the gland region ('toad no glands, toxin smeared'). Juvenile cane toads have a reduced overall toxin load compared to adults due to body size; however, they display the same bufadienolide compounds (Hayes et al. 2009), and should therefore still elicit a response from crows. These prey item treatments tested whether crows could: (a) distinguish between toxic toads, non-toxic toads, and native frogs; and, (b) identify the cues used to do this (visual or chemical). Comparing outcomes from treatments 2 (intact toad) and 3 (toads no glands) assessed whether crows rely on visual cues – specifically the enlarged parotoid glands, which contain most toxins – to identify toxic toads. Treatment 4 (toads no glands, toxin smeared) examined

whether chemical cues from toxins influenced behaviour in the absence of visual cues (treatments 3 and 4 could not be distinguished visually by a human observer). We sourced all prey items from opportunistic intact road-kill or humane euthanasia of live-captured animals. We used three species of native frogs: *Litoria pallida*, *Litoria coplandi* and *Litoria watjulumensis*, none of which are known to be toxic to crows. Snout-vent length (SVL) was measured for all prey items to the nearest 1 mm using callipers.

At each site we established 1–6 feeding stations, depending on crow abundance. Low abundance sites (such as roadside parking areas with 1–3 conspicuous crows, $n=9$) had a single feeding station with one randomly selected treatment. High abundance sites (e.g. landfills and campgrounds with ~50–200 crows, $n=15$) had 4–6 feeding stations set up simultaneously. At these high abundance sites, different treatments would be set up per feeding station, with no more than two stations with the same treatment at a given site. We treated each station as an independent replicate, assuming it was unlikely the same crow visited multiple stations given the large number of crows present and their tendency to aggregate and remain at specific stations.

In total, we set up 153 feeding stations across the four treatment groups: native frog ($n=29$); intact toad ($n=45$); toad with glands excised ($n=39$); toad no glands, toxin smeared ($n=40$). We obtained data from 49 feeding station trials where the outcome could be clearly determined, comprising the following treatments: native frog ($n=6$), intact toad ($n=18$), toad no glands ($n=13$), toad no glands, toxic smeared ($n=12$) (S1 Methods). We excluded 104 trials either because crows did not physically interact with the prey item ($n=56$) or because crows removed the prey item and we could not clearly determine the outcome ($n=48$). In several cases where no prey remained, we could use camera footage to determine that the entire prey had been consumed. When there were prey remnants, we classified consumption into 'toxic' and 'non-toxic' categories based on which prey parts were eaten. Non-toxic categories included leg muscles, arm muscles, and gut (internal organs); toxic categories included dorsal skin, head (not inclusive of the parotoid glands), and parotoid glands (Hayes et al. 2009). In 84% of trials, consumption of leg muscles, arm muscles, and gut contents was linked (all three either consumed or not). Similarly, consumption of dorsal skin, head and parotoid glands was linked in 84% of cases. Consumption outcomes were scored by a single observer (SD). Blind scoring was not possible given the visually distinct differences between treatments (e.g. the presence or absence of parotoid glands); however, this is unlikely to introduce bias as consumption of each body part was a visually unambiguous, binary outcome.

Analysis

We used consumption of toxic regions or not (coded as a 0/1 binary variable) as the response variable in our analyses. Our data suffered from complete separation, whereby some treatments contained all ones or zeros. Under these conditions, maximum likelihood estimates suffer from poor precision, so we fitted regression models in a Bayesian framework using weakly informative priors to regularise parameter estimates.

In addition to the prey item treatments, we included prey item size (SVL) and the length of time in years that crows had been exposed to cane toads as predictors in the regression model. We included prey size to control for potential differences in the size of adult native frogs and juvenile cane toads that might influence prey consumption. We first tested for size differences between these two groups using a Mann-Whitney U sum rank test. We estimated the length of time that crows at each site had been exposed to cane toads from publicly available maps with records of cane toad spread held by the Department of Biodiversity, Conservation and Attractions, Western Australia government and publications identifying specific invasion locations (e.g. Doody et al. 2018; DBCA 2025). We included cane toad exposure time as a predictor to determine if consumption patterns had changed over time as crows adapted to cane toad presence.

We used the package *brms* (Bürkner 2017) in R (R Core Team 2026) to fit Bayesian binomial regression models on the logit scale with consumption of toxic regions or not as the response variable, and prey item treatment (native frog, intact toad, toad no glands, and toad no glands, toxin smeared), snout-vent length, and years of toad exposure as fixed effect predictors, and site location as a random effect, allowing for non-independence among trials conducted at

the same site. We included weakly informative priors to regularise estimates, specifying priors for all fixed effects as normally distributed with mean zero and standard deviation two on the logit scale.

Results

Cane toads and native frogs did not differ significantly in SVL (cane toads: 57.7 ± 11.0 mm, native frogs: 54.9 ± 9.8 mm; mean $\pm$ SD; Mann-Whitney $U p = 0.1439$; Fig. S3). and there was also no clear effect of either size or length of exposure to cane toads on the probability of consuming toxic regions (Tables S2, S3; Fig. S4).

Crows entirely consumed all native frogs (6/6) and all cane toads that had their parotoid glands removed (toad no gland, 13/13). In 89% (16/18) of trials involving intact cane toads, crows avoided toxic regions while consuming non-toxic regions. In the two trials where toxic regions were consumed, the entire toad was consumed in one trial and only part of the dorsal skin was consumed in the remaining trial. In trials where cane toad glands were removed and toxin reapplied (toad no glands, toxin smeared) crows consumed toxic regions in 58% of trials (7/12).

These outcomes and their associated uncertainty intervals, estimated from the regression model, indicated a high probability that crows consumed putative toxic regions of the prey item in treatments involving native frogs and cane toads with glands removed (Fig. 1). The probability that crows consumed the toxic regions of intact toads relative to the native frog baseline was significantly lower (Figs. 1 and 2). The treatment involving gland removal with toxin reapplication resulted in an intermediate outcome: approximately half of the crows consumed toxic regions while half did not (Figs. 1 and 2).

Fig. 1 Posterior estimates of the probability that toxic regions of a prey item were consumed from the Bayesian regression model per treatment group (native frog, $n=6$; intact cane toad, $n=18$; cane toad with glands removed, $n=13$; and cane toad with glands removed and toxin smeared, $n=12$). For native frogs, the allocated toxic region is non-toxic and represents the proportion of the prey item consumed. For the remaining treatments, each point represents the median posterior probability that the toxic region is consumed, and the bars represent 95% credible intervals. The sample size of each treatment group is given above the graph

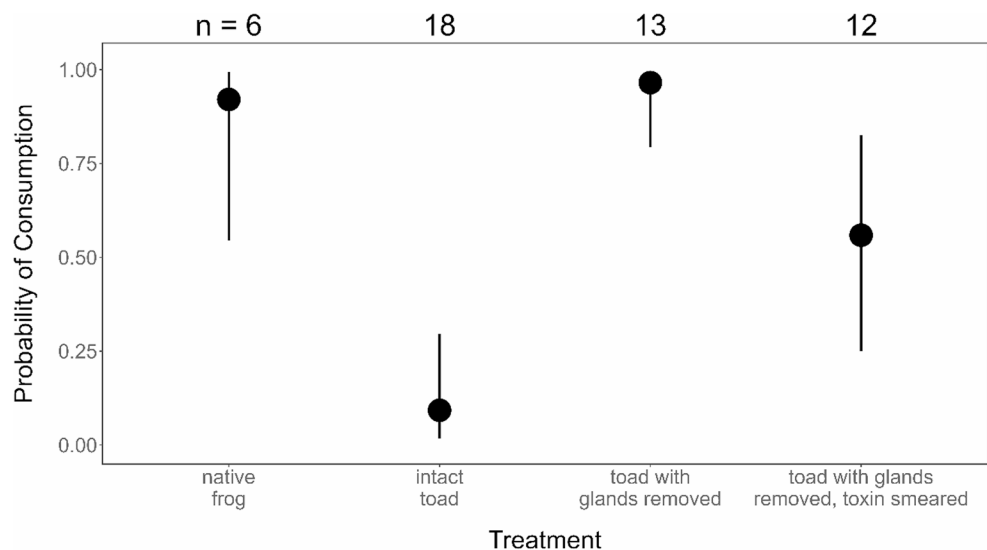
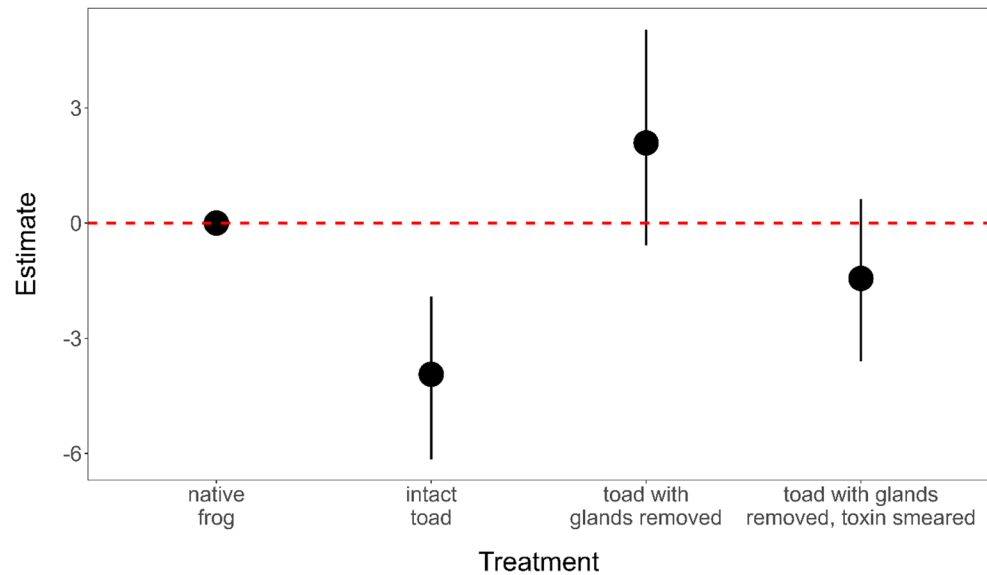


Fig. 2 Output of the Bayesian regression model showing differences in the log-odds of consuming toxic regions among treatment groups relative to native frogs, which were set as the reference baseline (red dashed line). Points represent the median posterior probability of the difference between native frogs and toad treatments in the probability of consuming toxic regions, and the bars represent 95% credible intervals



Discussion

Our study provides the first empirical evidence of safe consumption of toxic invasive cane toads (*R. marina*) by native Torresian crows (*C. orru*) across the East Kimberley, within 1–10 years post-invasion. Crows adjusted their consumption by manipulating prey, distinguishing between toxic and non-toxic prey, and modifying their behaviour when toxin cues were altered. This capacity to develop solutions to novel challenges strongly indicates behavioural flexibility—the ability to adjust behaviours to meet changing environmental conditions (Mazza and Šlipogor 2024).

Crow feeding behaviour was consistent regardless of time since cane toad invasion, suggesting both rapid emergence of behaviour to handle cane toads and persistence of these behaviours across populations and time. These adaptations allow crows not only to avoid lethal poisoning but also to exploit cane toads as an abundant food resource. Our findings add to growing evidence that some Australian native fauna are responding adaptively to cane toad invasion through behavioural means. We found that crows were able to differentiate between non-toxic native frogs and invasive toads, which has also been documented in native common planigales (*Planigale maculata*) (Webb et al. 2008) that have developed avoidance behaviours. This ability to discriminate between non-toxic native frogs and cane toads is suggestive that behaviour is not a pre-adaptive response to all anuran amphibians, but rather a targeted approach to cane toads. Native water rats (*Hydromys chrysogaster*) present behaviours similar to crows where they flip cane toads over to consume organs from the abdomen. Whether this behaviour is learnt or an ancestral safe-handling behaviour from handling all amphibians is unclear (Parrott et al. 2019). Other birds such as whistling kites (*Haliastur sphenurus*) and black kites

(*Milvus migrans*) also consume non-toxic toad organs through entry via the mouth, a behaviour we similarly observed in Torresian crows (Beckmann and Shine 2011). These comparisons suggest that native predators are increasingly displaying convergent behavioural strategies to overcome toxic cane toads, with crows representing a distinct example of acquired behaviours, rather than pre-adaptation, that enables a targeted and consistent approach to safely exploit toads.

Crow responses across treatments provide insight into the mechanisms underlying this behaviour. Crows consumed both native frogs and toads with toxin glands removed, while avoiding consuming toxic regions of intact toads, demonstrating an ability to distinguish between non-toxic and toxic amphibians. Prey size (SVL) did not clearly differ among treatments, ruling out size as a factor influencing consumption. Instead, the presence of large, visually distinct parotoid glands appears to be a key cue for identifying toxic prey. In the treatment where glands were removed but toxins smeared back onto the gland region, crows showed mixed responses; some crows began feeding as if the prey were non-toxic, then switched to avoidance despite the absence of the visual cue. This suggests that some crows could detect toxins, likely through chemical cues associated with taste aversion, although this response was weaker than visual cues alone. It is also possible that crows were detecting airborne chemical cues through olfaction, however, birds are traditionally considered to have lesser developed olfaction system compared to other taxa (Abankwah et al. 2020). There is some evidence that corvids can utilise olfaction cues (e.g. Harriman and Berger 1986; Rubene et al. 2019); however, visual cues appear to be the most significant mode of prey assessment (Santisteban et al. 2002; Rubene et al. 2019).

To our knowledge, no other predator is known to use the visual cue of parotoid glands to distinguish between toxic and

non-toxic amphibians, although this has not been explicitly tested. Similarly, while taste perception in birds is less studied than in mammals, evidence suggests that corvids can taste, and bitter compounds may play an important role in associating toxicity (Brinkman 2015; Destrez et al. 2022; Shoval et al. 2022). We have also observed frequent beak wiping in captive crows presented with toads for the first time (SD unpubl. data). Other taxa also use chemical cues to avoid cane toads including native frogs (Pizzatto and Shine 2009), fishes (to avoid cane toad tadpoles) (Caller and Brown 2013) and mammals (planigales) (Webb et al. 2008). Our results suggest that crows use multiple cues, with visual signals providing an initial assessment and chemical cues reinforcing avoidance – although chemical cue responses were mixed, with model estimates suggesting aversion, but not strongly. Similar strategies occur in other systems, such as predatory fish avoiding toxic corals, where visual cues can guide initial prey selection and chemical cues influence subsequent behaviour (Alino et al. 1992).

Our study provides clear evidence of behavioural flexibility in Torresian crows, enabling them to safely and efficiently exploit a novel, toxic prey item – the cane toad – soon after invasion. Crows selectively consumed non-toxic regions of cane toads and adjusted their behaviour in response to our treatments, demonstrating a sophisticated and flexible foraging strategy. This ability to rapidly modify behaviour in the face of a novel ecological threat highlights behavioural flexibility as a powerful mechanism for resilience among native predators. The emergence and persistence of this behaviour across multiple recently invaded crow populations suggests that such flexibility may play a critical role in mitigating the impacts of invasive species. By converting a potentially lethal prey item into a reliable food source, Torresian crows demonstrate how native species can adaptively respond to environmental change through behavioural means. Behavioural adaptation may also explain examples of cane toad resilience in other frog eating predators that are otherwise not necessarily resistant to toad toxins.

These findings provide opportunities for novel conservation strategies that support or enhance behavioural adaptation in vulnerable species, improving their chances of persistence in ecosystems under pressure from biological invasions. For example, as crows have the potential to learn from one another, crows that have encountered toads previously could be moved beyond the invasion front to teach others. As global biodiversity continues to decline due to invasive species and other anthropogenic pressures, understanding the conditions that promote behavioural adaptation could be key to enhancing species long-term survival. Our work provides a foundation for future research into the emergence, spread, and persistence of adaptive behaviours across populations – critical components for understanding resilience in rapidly changing environments.

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Author contribution SD: wrote the main manuscript text, conceptualisation of ideas, developed experimental design, carried out field work and data collection, undertook formal analysis, reviewing and editing of the manuscript.

MJW: conceptualisation of ideas, developed experimental design, carried out field work and data collection, undertook formal analysis, reviewing and editing of the manuscript.

JSD: conceptualisation of ideas, developed experimental design, reviewing and editing of the manuscript.

RPD: conceptualisation of ideas, developed experimental design, carried out field work and data collection, undertook formal analysis, reviewing and editing of the manuscript.

HF: carried out field work and data collection, conceptualisation of ideas, developed experimental design.

SC: conceptualisation of ideas, developed experimental design, carried out field work and data collection, undertook formal analysis, reviewing and editing of the manuscript.

All authors contributed to the manuscript and approved the final version.

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Data availability All data and code are included in the supplementary materials and are available at the Open Science Framework repository for this manuscript, <https://doi.org/10.17605/OSF.IO/5UXC2>.

Declarations

Ethics approval All applicable international, national, and/or institutional guidelines for the use of animals were followed. All animal experiments were conducted under Macquarie University Ethics 2019/016–7 and permit FO25000092 issued by the Department of Biodiversity, Conservation and Attractions (Western Australian government).

Competing interests The authors declare no competing interests.

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